Right-Triangle Trigonometry

Concept

Trigonometric ratios connect an acute angle with side lengths in a right triangle.

SOH CAH TOA

$\sin\theta = o/h$; $\cos\theta = a/h$; $\tan\theta = o/a$.

Label first

Hypotenuse is opposite 90°; o and a depend on the chosen angle.

Find an angle

Use inverse trig and degree mode for this unit.

Applications

Elevation/depression use a horizontal reference; include eye height.

Example 1

Find the opposite side when $\theta = 35°$ and the hypotenuse is 10 cm.

Solution

1. $\sin 35° = o/10$.

2. $o = 10\sin 35° \approx 5.74$ cm.

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Example 2

A ramp rises 1.2 m over a horizontal run of 8 m. Find its angle.

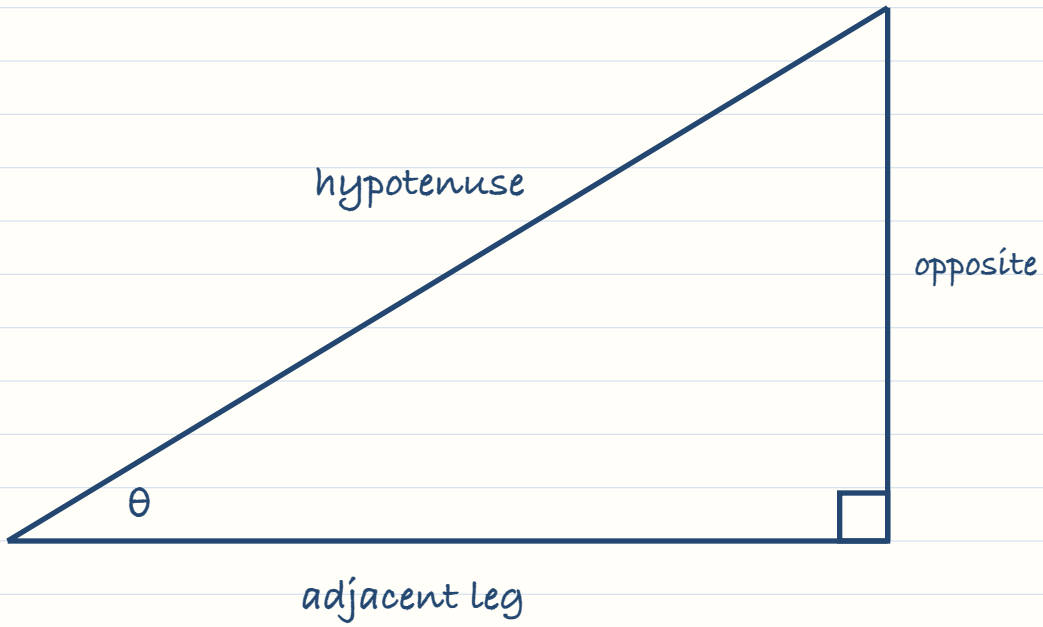
Solution

1. $\tan\theta = 1.2/8 = 0.15$.

2. $\theta = \tan^{-1}(0.15) \approx 8.53°$.

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Visual notes | label before calculating



The hypotenuse is opposite the right angle. Opposite and adjacent refer to the selected acute angle θ. A different selected angle changes those two labels.

Pythagoras: $a^2 + b^2 = c^2$. For right triangles, sin θ = opposite/hypotenuse, cos θ = adjacent/hypotenuse, and tan θ = opposite/adjacent.

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Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Find the opposite side for $\theta = 30°$ and hypotenuse 12.

2. Find θ when opposite $= 7$ and adjacent $= 9$.

3. Find hypotenuse when adjacent $= 5$ and $\theta = 60°$.

4. Why must you use horizontal distance, not the slope length, as adjacent in an elevation problem?

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Practice answers | check after trying

1. $12 \sin 30^\circ = 6$.

2. $\theta = \tan^{-1}(7/9) \approx 37.87^\circ$.

3. $h = 5/\cos 60^\circ = 10$.

4. The adjacent leg lies along the horizontal reference line; the slope length is the hypotenuse, so using it as adjacent changes the triangle ratio.